

# Task 2 Report

## Clustering Analysis of Embedding Space

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# INTRODUCTION

This report presents the results of Task 2, whose objective is to assess whether patient embeddings learned in an unsupervised manner naturally organise into clinically coherent groups.

While Task 1 focused on supervised prediction of clinical targets (PAM50 subtype, survival risk, and pathological stage), Task 2 addresses a complementary question: do the learned representations exhibit an intrinsic organisation that reflects clinically relevant categories without the use of supervision?

To answer this question, five clustering algorithms were applied to the four embedding levels (Levels 0 to 3). Clustering quality was evaluated using internal metrics (Silhouette score, Davies–Bouldin index, and noise proportion for HDBSCAN), external metrics (Adjusted Rand Index and Normalized Mutual Information with respect to PAM50 labels), and chi-square tests assessing associations between cluster assignments and clinical variables.

From a clinical perspective, the selected variables are particularly relevant. PAM50 molecular subtypes are widely used in breast cancer management to guide therapeutic decisions and estimate prognosis. Survival status provides direct prognostic information, while pathological stage and TNM classification characterise disease extent and tumour progression. Using these variables as references therefore provides an indirect measure of the clinical relevance of the discovered structures.

From a methodological perspective, the incorporation of protein–protein interaction (PPI) networks as an inductive bias constitutes a distinctive aspect of this project. The underlying assumption is that integrating prior biological knowledge during representation learning, either through graph regularisation (Level 2 (GR-DAE)) or through graph-based message passing within a GNN encoder (Level 3 (GS-DAE)), encourages the emergence of biologically meaningful and clinically informative structures within the embedding space.

The following hypotheses are formulated:

- **Level 2** (GR-DAE) will produce more clinically coherent clusters than Levels 0 (AE) and 1 (DAE) by constraining the latent space toward a biologically structured representation.
- **Level 3** (GS-DAE) will exhibit the strongest internal geometric structure. However, this organisation may primarily reflect graph-induced smoothing effects rather than clinically meaningful subtype separation.

- **HDBSCAN**, due to its ability to adapt to clusters of varying density and shape, will identify the most geometrically coherent structures within Level 3 (GS-DAE) embeddings compared with fixed-k clustering approaches.
- **Spectral Clustering** may be poorly suited to Level 3 (GS-DAE) embeddings because the fragmented geometry of this representation could prevent the construction of a coherent neighbourhood graph, which constitutes a fundamental assumption of spectral methods.

This report presents the experimental protocol, clustering analyses, statistical comparisons between representation levels and algorithms, and the clinical interpretation of the identified structures. A clinically grounded discussion and perspectives for the continuation of the project are also provided.

# 1 EXPERIMENTAL PROTOCOL

## 1.1 CLUSTERING ALGORITHMS

Five clustering algorithms were applied to each of the four embedding levels (Levels 0 to 3), covering a broad range of clustering paradigms:

- **K-Means (centroid-based):** partitions the data into a fixed number of clusters by minimising the within-cluster variance.
- **Agglomerative Hierarchical Clustering (hierarchical):** builds a hierarchy of clusters through successive merges using Ward's linkage criterion.
- **HDBSCAN (density-based):** identifies clusters of arbitrary shape and automatically handles noise by assigning outlier samples to a dedicated noise class (labelled as -1).
- **Spectral Clustering (graph-based):** uses the eigenvectors of a similarity matrix to project the data into a lower-dimensional space before clustering.
- **Gaussian Mixture Models (GMM) (probabilistic):** model the data as a mixture of Gaussian distributions and allow soft probabilistic assignment of patients to clusters.

These algorithms were selected to capture different types of latent structures and to evaluate the robustness of the observed clustering patterns across multiple methodological assumptions.

## 1.2 EVALUATION METRICS

### Internal metrics

The intrinsic quality of the clustering solutions was assessed using three complementary internal metrics:

- **Silhouette score:** measures how similar each patient is to its own cluster compared to neighbouring clusters. Values range from -1 to 1, with higher values indicating better cluster separation.
- **Davies–Bouldin index (DBI):** quantifies the average similarity between each cluster and its most similar neighbouring cluster. Lower values indicate more compact and better-separated clusters.

- **Noise proportion:** specific to HDBSCAN, corresponding to the percentage of patients classified as outliers (label -1).

### External metrics

To evaluate the agreement between unsupervised clusters and known molecular subtypes, clustering assignments were compared with PAM50 labels using:

- **Adjusted Rand Index (ARI):** measures the similarity between cluster assignments and ground truth labels while correcting for random agreement.
- **Normalised Mutual Information (NMI):** quantifies the amount of shared information between cluster assignments and reference labels.

### Clinical validation

To assess the clinical relevance of the discovered structures, chi-square tests of independence were performed between cluster assignments and several clinical variables:

- PAM50 subtype,
- survival group,
- pathological stage,
- T\_stage,
- N\_stage,
- M\_stage.

## 1.3 STATISTICAL COMPARISONS

To evaluate whether observed differences between representation levels and clustering algorithms were statistically significant, paired t-tests were applied to clustering metrics obtained across repeated runs of stochastic algorithms.

The analyses focused on:

- **internal metrics** (Silhouette score and Davies–Bouldin index),
- **external metrics** (Adjusted Rand Index).

A significance threshold of  $p < 0.05$  was used throughout the study.

Two categories of comparisons were considered:

- between representation levels: Level 2 (GR-DAE) vs Level 0 (AE), Level 2 (GR-DAE) vs Level 1 (DAE), Level 3 (GS-DAE) vs Level 2 (GR-DAE), and Level 3 (GS-DAE) vs Level 0 (AE).
- between clustering algorithms: HDBSCAN vs K-Means, HDBSCAN vs Spectral Clustering, and GMM vs K-Means on Level 3 (GS-DAE) embeddings.

## 1.4 REPEATED EXPERIMENTS

To ensure robustness and estimate variability, stochastic algorithms (K-Means, Gaussian Mixture Models, and Spectral Clustering) were repeated 10 times using different random seeds. Results are reported as mean  $\pm$  standard deviation across runs.

Deterministic algorithms (Agglomerative Clustering and HDBSCAN) were executed once, as they do not rely on random initialisation.

The following sections present the selection of clustering parameters, the evaluation of clustering quality, the analysis of clinical associations, and a discussion of the observed structures.



## 2 CHOICE OF NUMBER OF CLUSTERS (K)

To compare clustering results across algorithms and representation levels, a fixed number of clusters ( $k$ ) was required for algorithms relying on an explicit specification of the number of clusters, namely K-Means, Agglomerative Clustering, Spectral Clustering and Gaussian Mixture Models (GMM). The choice of ( $k$ ) was guided by two complementary criteria: the elbow method for K-Means and the Bayesian Information Criterion (BIC) for GMM. Additional analyses were performed for ( $k=3$ ), ( $k=4$ ), and ( $k=5$ ) on the most relevant representation levels to validate the final selection.

### 2.1 ELBOW METHOD (K-MEANS)

The elbow method was applied to K-Means for ( $k \in \{2, \dots, 8\}$ ). The within-cluster sum of squares (inertia) was computed for each representation level.

As shown in Figure 1 (DAE), all levels exhibited a decrease in inertia as the number of clusters increased, as expected. A visible change in slope was consistently observed around ( $k=4$ ), suggesting that four clusters provide a reasonable trade-off between cluster compactness and model complexity.

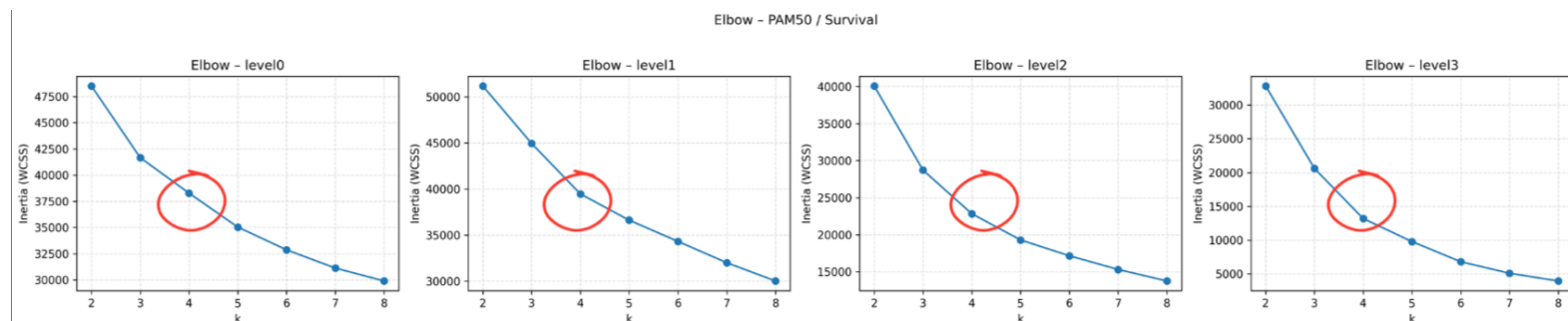


Figure 1 : An elbow is visible around  $k=4$  at all levels.

### 2.2 BAYESIAN INFORMATION CRITERION (GMM)

For Gaussian Mixture Models, the Bayesian Information Criterion (BIC) was computed over the same range of cluster numbers.

The optimal number of components varied depending on the representation level:

- **Level 0:** ( $k=6$ )

- **Level 1:** ( $k=5$ )
- **Level 2:** ( $k=4$ )
- **Level 3:** ( $k=3$ )

This variability suggests that the intrinsic geometry of the embedding space changes substantially across representation levels. Nevertheless, using a different value of ( $k$ ) for each level would make direct comparisons between algorithms and representations more difficult.

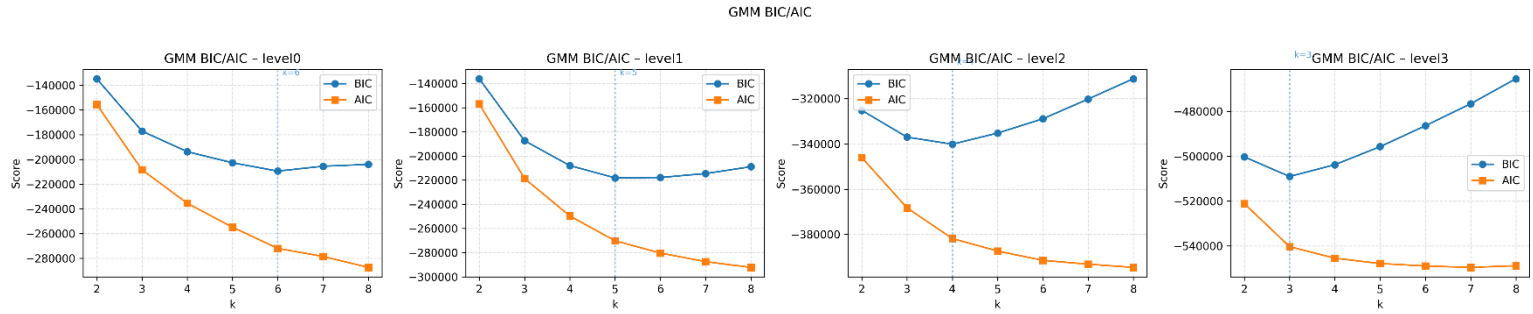


Figure 2 : The BIC suggests  $k=4$  for Level 2, but  $k=3$  for Level 3.

## 2.3 ADDITIONAL COMPARISON OF ( $k=3$ ), ( $k=4$ ), AND ( $k=5$ )

To further assess the sensitivity of clustering performance to the choice of ( $k$ ), additional experiments were performed using ( $k=3$ ), ( $k=4$ ), and ( $k=5$ ) on the representation levels that exhibited the strongest clustering tendencies in preliminary analyses (Levels 2 (GR-DAE) and 3 (GS-DAE)).

The results indicated that:

- For **Level 2** (GR-DAE), K-Means achieved its highest silhouette score for ( $k=5$ ), although the improvement over ( $k=4$ ) remained modest.
- For **Level 3** (GS-DAE), HDBSCAN substantially outperformed fixed- $(k)$  methods by automatically identifying around 30 clusters with a silhouette score close to 0.80.
- For **Levels 0** (AE) and **1** (DAE), silhouette scores remained low regardless of the value of ( $k$ ), indicating weak intrinsic cluster structure.

A summary of the best clustering configurations is provided below.

| Level          | Algorithm                       | Best k | Silhouette | Comment                        |
|----------------|---------------------------------|--------|------------|--------------------------------|
| <b>Level 0</b> | K-Means<br>Agglomerative<br>GMM | 2      | ~0.25      | Very low structure             |
| <b>Level 1</b> | HDBSCAN                         | Auto   | 0.24       | Very low structure             |
| <b>Level 2</b> | K-Means                         | 5      | 0.46       | Best for this level            |
| <b>Level 3</b> | HDBSCAN                         | Auto   | 0.80       | Best structure (oversmoothing) |

## 2.4 FINAL CHOICE OF $k = 4$

Based on the previous analyses, ( $k=4$ ) was selected as the reference value for the main clustering experiments for several reasons:

- it was consistently supported by the elbow criterion across all representation levels;
- it provided a reasonable compromise between clustering compactness and separation, as reflected by the internal clustering metrics presented in the following sections;
- it enabled direct comparisons across clustering algorithms and embedding representations;
- it remained close to the optimal configurations identified by alternative criteria.

Although K-Means achieved slightly better performance on Level 2 (GR-DAE) with ( $k=5$ ), the difference compared with ( $k=4$ ) remained limited (silhouette score: 0.46 versus 0.44). Similarly, HDBSCAN achieved substantially higher silhouette scores on Level 3 (GS-DAE) using an automatically determined number of clusters, but these clusters likely reflect the highly regular geometry induced by graph propagation and potential over-smoothing effects.

Consequently, ( $k=4$ ) was adopted as the common number of clusters for the main analyses, while HDBSCAN results were analysed separately as a complementary characterisation of the intrinsic geometry of the Level 3 (GS-DAE) embedding space.

## 3 INTERNAL CLUSTERING METRICS

### 3.1 OVERVIEW

Using the evaluation metrics introduced in Section 1.2, the intrinsic quality of the clustering solutions was assessed for each clustering algorithm and each representation level independently of clinical labels.

For algorithms requiring an explicit specification of the number of clusters (K-Means, Agglomerative Clustering, Spectral Clustering, and Gaussian Mixture Models), the common value ( $k=4$ ) selected in Section 2 was used. For HDBSCAN, the number of clusters was determined automatically by the algorithm.

### 3.2 COMPLETE RESULTS

The table below summarises the Silhouette scores and Davies–Bouldin indices obtained for all clustering algorithms across the four representation levels.

| Target         | Algorithm | Level  | k | Silhouette | Silhouette_std | Davies_Bouldin |
|----------------|-----------|--------|---|------------|----------------|----------------|
| PAM50/Survival | KMEANS    | level0 | 2 | 0.2530     | 0.0            | 1.6489         |
| PAM50/Survival | KMEANS    | level0 | 3 | 0.2326     | 0.0            | 1.7486         |
| PAM50/Survival | KMEANS    | level0 | 4 | 0.2326     | 2.7756e-17     | 1.6745         |
| PAM50/Survival | KMEANS    | level0 | 5 | 0.2210     | 8.4697e-05     | 1.6448         |
| PAM50/Survival | KMEANS    | level0 | 6 | 0.2211     | 0.0027         | 1.5317         |
| PAM50/Survival | KMEANS    | level0 | 7 | 0.2216     | 0.0017         | 1.5809         |
| PAM50/Survival | KMEANS    | level0 | 8 | 0.2165     | 0.0187         | 1.5402         |
| PAM50/Survival | KMEANS    | level1 | 2 | 0.2242     | 2.7756e-17     | 1.8374         |
| PAM50/Survival | KMEANS    | level1 | 3 | 0.2167     | 0.0001         | 1.7943         |
| PAM50/Survival | KMEANS    | level1 | 4 | 0.2158     | 0.0007         | 1.4385         |
| PAM50/Survival | KMEANS    | level1 | 5 | 0.2104     | 0.01293        | 1.4760         |
| PAM50/Survival | KMEANS    | level1 | 6 | 0.2085     | 0.0040         | 1.5099         |
| PAM50/Survival | KMEANS    | level1 | 7 | 0.21160    | 0.0011         | 1.5415         |
| PAM50/Survival | KMEANS    | level1 | 8 | 0.2132     | 0.0012         | 1.4535         |
| PAM50/Survival | KMEANS    | level2 | 2 | 0.3570     | 0.0            | 1.2470         |
| PAM50/Survival | KMEANS    | level2 | 3 | 0.4091     | 5.5511e-17     | 0.9757         |
| PAM50/Survival | KMEANS    | level2 | 4 | 0.4412     | 2.5613e-05     | 0.9198         |
| PAM50/Survival | KMEANS    | level2 | 5 | 0.4572     | 7.8852e-05     | 0.8910         |
| PAM50/Survival | KMEANS    | level2 | 6 | 0.43508    | 0.0253         | 0.8895         |
| PAM50/Survival | KMEANS    | level2 | 7 | 0.4122     | 0.0108         | 0.9008         |
| PAM50/Survival | KMEANS    | level2 | 8 | 0.4041     | 0.0027         | 0.9367         |
| PAM50/Survival | KMEANS    | level3 | 2 | 0.4391     | 0.0            | 0.9277         |

|                |          |        |      |        |            |         |
|----------------|----------|--------|------|--------|------------|---------|
| PAM50/Survival | KMEANS   | level3 | 3    | 0.4557 | 5.5511e-17 | 0.8735  |
| PAM50/Survival | KMEANS   | level3 | 4    | 0.5092 | 0.0        | 0.7343  |
| PAM50/Survival | KMEANS   | level3 | 5    | 0.5383 | 0.01272    | 0.6633  |
| PAM50/Survival | KMEANS   | level3 | 6    | 0.5791 | 0.0        | 0.5889  |
| PAM50/Survival | KMEANS   | level3 | 7    | 0.5850 | 0.01002    | 0.5546  |
| PAM50/Survival | KMEANS   | level3 | 8    | 0.5709 | 0.0041     | 0.5542  |
| PAM50/Survival | AGG      | level0 | 2    | 0.2514 | 0.0        | 1.6428  |
| PAM50/Survival | AGG      | level0 | 3    | 0.2267 | 0.0        | 1.8171  |
| PAM50/Survival | AGG      | level0 | 4    | 0.2065 | 0.0        | 1.7789  |
| PAM50/Survival | AGG      | level0 | 5    | 0.1793 | 0.0        | 1.8265  |
| PAM50/Survival | AGG      | level0 | 6    | 0.1820 | 0.0        | 1.5349  |
| PAM50/Survival | AGG      | level0 | 7    | 0.1914 | 0.0        | 1.7610  |
| PAM50/Survival | AGG      | level0 | 8    | 0.2006 | 0.0        | 1.7741  |
| PAM50/Survival | AGG      | level1 | 2    | 0.2242 | 0.0        | 1.8373  |
| PAM50/Survival | AGG      | level1 | 3    | 0.2103 | 0.0        | 1.8099  |
| PAM50/Survival | AGG      | level1 | 4    | 0.2088 | 0.0        | 1.3543  |
| PAM50/Survival | AGG      | level1 | 5    | 0.2120 | 0.0        | 1.4548  |
| PAM50/Survival | AGG      | level1 | 6    | 0.1868 | 0.0        | 1.5811  |
| PAM50/Survival | AGG      | level1 | 7    | 0.2009 | 0.0        | 1.5947  |
| PAM50/Survival | AGG      | level1 | 8    | 0.2013 | 0.0        | 1.4022  |
| PAM50/Survival | AGG      | level2 | 2    | 0.3482 | 0.0        | 1.3336  |
| PAM50/Survival | AGG      | level2 | 3    | 0.3962 | 0.0        | 1.0567  |
| PAM50/Survival | AGG      | level2 | 4    | 0.4252 | 0.0        | 0.9690  |
| PAM50/Survival | AGG      | level2 | 5    | 0.4439 | 0.0        | 0.9520  |
| PAM50/Survival | AGG      | level2 | 6    | 0.4461 | 0.0        | 0.8023  |
| PAM50/Survival | AGG      | level2 | 7    | 0.3944 | 0.0        | 0.8963  |
| PAM50/Survival | AGG      | level2 | 8    | 0.3765 | 0.0        | 0.9823  |
| PAM50/Survival | AGG      | level3 | 2    | 0.4840 | 0.0        | 0.8389  |
| PAM50/Survival | AGG      | level3 | 3    | 0.4443 | 0.0        | 0.9037  |
| PAM50/Survival | AGG      | level3 | 4    | 0.4987 | 0.0        | 0.7724  |
| PAM50/Survival | AGG      | level3 | 5    | 0.5426 | 0.0        | 0.6541  |
| PAM50/Survival | AGG      | level3 | 6    | 0.5791 | 0.0        | 0.5889  |
| PAM50/Survival | AGG      | level3 | 7    | 0.6003 | 0.0        | 0.4887  |
| PAM50/Survival | AGG      | level3 | 8    | 0.5776 | 0.0        | 0.5894  |
| PAM50/Survival | HDBSCAN  | level0 | auto | 0.2413 | 0.0        | 1.2309  |
| PAM50/Survival | HDBSCAN  | level1 | auto | 0.2377 | 0.0        | 1.41667 |
| PAM50/Survival | HDBSCAN  | level2 | auto | 0.2992 | 0.0        | 0.9622  |
| PAM50/Survival | HDBSCAN  | level3 | auto | 0.8009 | 0.0        | 0.2829  |
| PAM50/Survival | SPECTRAL | level0 | 2    | 0.1301 | 0.0482     | 2.0171  |
| PAM50/Survival | SPECTRAL | level0 | 3    | 0.0787 | 0.0358     | 2.3672  |
| PAM50/Survival | SPECTRAL | level0 | 4    | 0.0338 | 0.0461     | 2.3556  |
| PAM50/Survival | SPECTRAL | level0 | 5    | 0.0731 | 0.0303     | 2.2173  |
| PAM50/Survival | SPECTRAL | level0 | 6    | 0.0363 | 0.0409     | 2.2212  |
| PAM50/Survival | SPECTRAL | level0 | 7    | 0.0736 | 0.0346     | 2.1612  |
| PAM50/Survival | SPECTRAL | level0 | 8    | 0.0706 | 0.0350     | 2.1311  |
| PAM50/Survival | SPECTRAL | level1 | 2    | 0.1314 | 0.0209     | 1.9819  |
| PAM50/Survival | SPECTRAL | level1 | 3    | 0.0719 | 0.0397     | 2.0502  |
| PAM50/Survival | SPECTRAL | level1 | 4    | 0.0613 | 0.0476     | 2.4525  |
| PAM50/Survival | SPECTRAL | level1 | 5    | 0.0679 | 0.0339     | 2.2643  |
| PAM50/Survival | SPECTRAL | level1 | 6    | 0.0608 | 0.0275     | 2.2287  |

|                |          |        |   |         |            |        |
|----------------|----------|--------|---|---------|------------|--------|
| PAM50/Survival | SPECTRAL | level1 | 7 | 0.0668  | 0.0474     | 2.1833 |
| PAM50/Survival | SPECTRAL | level1 | 8 | 0.0772  | 0.0332     | 2.1365 |
| PAM50/Survival | SPECTRAL | level2 | 2 | 0.2171  | 0.0479     | 1.1696 |
| PAM50/Survival | SPECTRAL | level2 | 3 | 0.0625  | 0.0876     | 1.4936 |
| PAM50/Survival | SPECTRAL | level2 | 4 | 0.0715  | 0.0411     | 1.9799 |
| PAM50/Survival | SPECTRAL | level2 | 5 | 0.0530  | 0.0675     | 1.6159 |
| PAM50/Survival | SPECTRAL | level2 | 6 | 0.0561  | 0.0526     | 1.8237 |
| PAM50/Survival | SPECTRAL | level2 | 7 | 0.0602  | 0.0522     | 1.6363 |
| PAM50/Survival | SPECTRAL | level2 | 8 | 0.0715  | 0.0406     | 1.8331 |
| PAM50/Survival | SPECTRAL | level3 | 2 | -0.1887 | 0.0492     | 2.0808 |
| PAM50/Survival | SPECTRAL | level3 | 3 | -0.2184 | 0.1262     | 4.0172 |
| PAM50/Survival | SPECTRAL | level3 | 4 | -0.2649 | 0.0789     | 5.4459 |
| PAM50/Survival | SPECTRAL | level3 | 5 | -0.3464 | 0.0456     | 3.2764 |
| PAM50/Survival | SPECTRAL | level3 | 6 | -0.3102 | 0.0585     | 3.9870 |
| PAM50/Survival | SPECTRAL | level3 | 7 | -0.3204 | 0.0715     | 4.1596 |
| PAM50/Survival | SPECTRAL | level3 | 8 | -0.3598 | 0.0489     | 3.3555 |
| PAM50/Survival | GMM      | level0 | 2 | 0.2518  | 0.0006     | 1.6440 |
| PAM50/Survival | GMM      | level0 | 3 | 0.2305  | 2.7756e-17 | 1.7583 |
| PAM50/Survival | GMM      | level0 | 4 | 0.2149  | 0.0026     | 1.8538 |
| PAM50/Survival | GMM      | level0 | 5 | 0.2173  | 0.0069     | 1.8223 |
| PAM50/Survival | GMM      | level0 | 6 | 0.2109  | 0.0066     | 1.9608 |
| PAM50/Survival | GMM      | level0 | 7 | 0.2102  | 0.0070     | 1.8901 |
| PAM50/Survival | GMM      | level0 | 8 | 0.1833  | 0.0184     | 2.0827 |
| PAM50/Survival | GMM      | level1 | 2 | 0.2171  | 0.0212     | 1.8947 |
| PAM50/Survival | GMM      | level1 | 3 | 0.2132  | 0.0006     | 1.8070 |
| PAM50/Survival | GMM      | level1 | 4 | 0.2042  | 0.0052     | 2.0818 |
| PAM50/Survival | GMM      | level1 | 5 | 0.2098  | 0.0026     | 1.9971 |
| PAM50/Survival | GMM      | level1 | 6 | 0.1822  | 0.0209     | 2.1496 |
| PAM50/Survival | GMM      | level1 | 7 | 0.1774  | 0.0172     | 2.1528 |
| PAM50/Survival | GMM      | level1 | 8 | 0.1813  | 0.0208     | 2.0611 |
| PAM50/Survival | GMM      | level2 | 2 | 0.3498  | 0.0016     | 1.2871 |
| PAM50/Survival | GMM      | level2 | 3 | 0.4064  | 0.0001     | 0.9824 |
| PAM50/Survival | GMM      | level2 | 4 | 0.4373  | 5.8503e-05 | 0.9249 |
| PAM50/Survival | GMM      | level2 | 5 | 0.4544  | 0.0004     | 0.8980 |
| PAM50/Survival | GMM      | level2 | 6 | 0.4048  | 0.0014     | 0.9872 |
| PAM50/Survival | GMM      | level2 | 7 | 0.3677  | 0.0280     | 1.0627 |
| PAM50/Survival | GMM      | level2 | 8 | 0.3426  | 0.0244     | 1.1038 |
| PAM50/Survival | GMM      | level3 | 2 | 0.3953  | 0.0669     | 1.1192 |
| PAM50/Survival | GMM      | level3 | 3 | 0.4593  | 5.5511e-17 | 0.8022 |
| PAM50/Survival | GMM      | level3 | 4 | 0.5089  | 0.0006     | 0.7318 |
| PAM50/Survival | GMM      | level3 | 5 | 0.5287  | 0.0106     | 0.7093 |
| PAM50/Survival | GMM      | level3 | 6 | 0.5270  | 0.0354     | 0.6930 |
| PAM50/Survival | GMM      | level3 | 7 | 0.5375  | 0.0082     | 0.6737 |
| PAM50/Survival | GMM      | level3 | 8 | 0.5511  | 0.0110     | 0.6768 |

To facilitate comparisons across methods and representation levels, the main results are also illustrated in Figure 3.

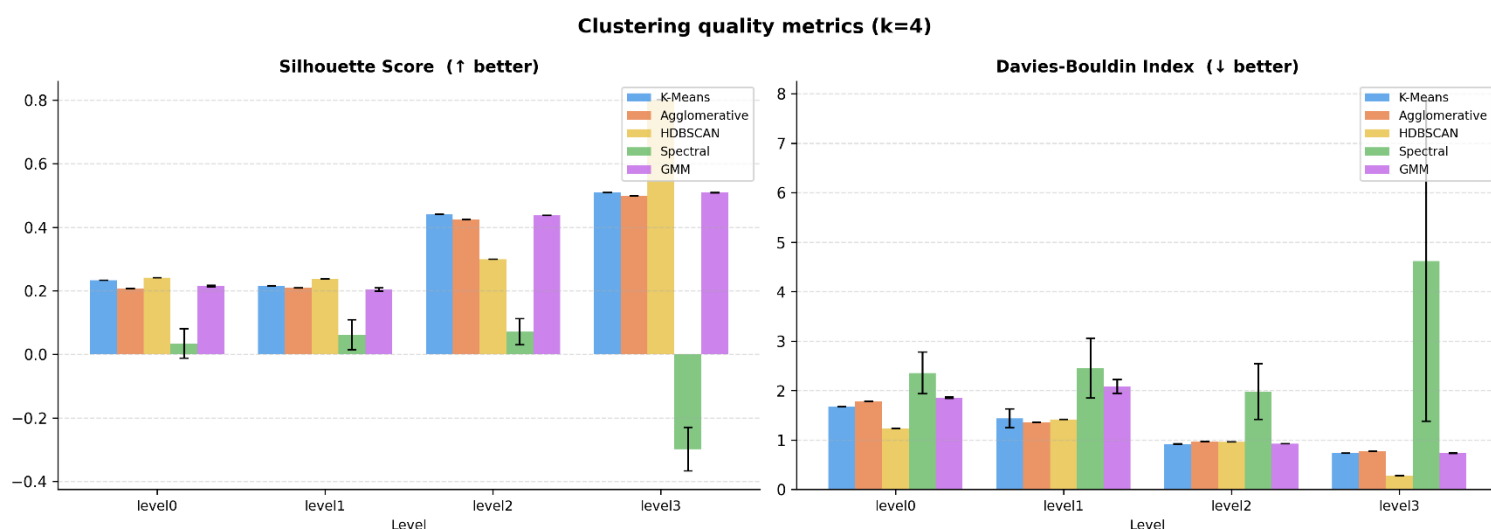


Figure 3 : HDBSCAN at Level 3 achieves the highest silhouette score (0.80).

Furthermore, as HDBSCAN explicitly models outlier samples, the proportion of patients classified as noise was evaluated for this algorithm across all representation levels. The table below details these noise proportions alongside the automatically determined number of clusters.

| Level   | Number of clusters | Noise (%) |
|---------|--------------------|-----------|
| Level 0 | 4                  | 3.3%      |
| Level 1 | 3                  | 2.4%      |
| Level 2 | 27                 | 28.4%     |
| Level 3 | 30                 | 0.2%      |

### 3.3 GLOBAL TRENDS ACROSS REPRESENTATION LEVELS

A progressive improvement in clustering quality is observed across representation levels.

Levels 0 (AE) and 1 (DAE) exhibit relatively low Silhouette scores, generally below 0.25, together with comparatively high Davies–Bouldin indices. These results indicate weak cluster organisation and substantial overlap between groups, suggesting that the embedding spaces learned by the standard autoencoder and denoising autoencoder contain limited intrinsic structure.

Level 2 (GR-DAE) shows a moderate improvement in both metrics, indicating that the incorporation of PPI-based regularisation introduces additional organisation into the embedding space.



The most pronounced improvement is observed for Level 3 (GS-DAE), which consistently achieves the highest Silhouette scores (0.80) and the lowest Davies–Bouldin indices (0.28) across most clustering algorithms. This suggests the presence of highly compact and well-separated regions within the latent space.

### **3.4 COMPARISON ACROSS CLUSTERING ALGORITHMS**

Among the fixed-(k) algorithms, GMM and K-Means generally achieve the best internal clustering performance, particularly on Level 3 (GS-DAE) where their Silhouette scores become substantially higher than those observed on lower representation levels.

Agglomerative clustering produces comparable results but generally remains slightly below K-Means and GMM in terms of cluster compactness and separation.

In contrast, Spectral Clustering performs considerably worse across all levels, often yielding Silhouette scores close to zero or even negative values. This behaviour suggests that the assumptions underlying spectral methods may not be compatible with the geometry of the learned embedding spaces. A detailed analysis of this phenomenon is provided in Section 7.

HDBSCAN exhibits a markedly different behaviour. By adapting the number of clusters to local density variations, it identifies a much finer partitioning of the embedding space than fixed-(k) methods. The characteristics of these clusters, including their stability, size distribution, and noise proportion, are analysed in detail in the following section.

### **3.5 BEST PERFORMING CONFIGURATION**

The best overall internal clustering performance is obtained with HDBSCAN applied to Level 3 (GS-DAE) embeddings. On this representation level, HDBSCAN identifies approximately 30 clusters while classifying only 0.2% of patients as noise.

These results indicate that the Level 3 (GS-DAE) embedding space is composed of numerous dense and well-separated regions rather than a small number of broad patient groups. Whether this organisation reflects biologically meaningful patient stratification or a regularisation effect induced by graph propagation is investigated in the subsequent analyses.

### 3.6 STATISTICAL COMPARISON OF INTERNAL METRICS

To formally assess whether the observed differences in clustering quality were statistically significant, paired t-tests were performed on the silhouette scores obtained over the 10 repeated runs of the stochastic clustering algorithms.

The comparisons between representation levels using K-Means revealed statistically significant differences in all cases ( $p < 0.001$ ). In particular, Level 2 (GR-DAE) significantly outperformed Levels 0 (AE) and 1 (DAE), confirming that the introduction of graph regularisation improves the geometric organisation of the embedding space. Furthermore, Level 3 (GS-DAE) significantly outperformed Level 2 (GR-DAE), indicating that graph propagation through the GNN encoder produces a substantially more structured latent space.

On Level 3 (GS-DAE), the comparison between GMM and K-Means did not reveal a statistically significant difference in silhouette score ( $p = 0.34$ ), suggesting that both algorithms achieve comparable clustering quality on this representation level.

Statistical comparisons involving HDBSCAN were not performed because HDBSCAN is deterministic and does not rely on random initialisation. Consequently, repeated runs yield identical clustering solutions, making paired statistical testing inappropriate.

Overall, these statistical analyses confirm that the improvements in clustering quality observed from Levels 0 to 3 are robust and unlikely to result from random variation.

## 4 HDBSCAN TUNING AND CLUSTER REGULARITY

### 4.1 STABILITY OF HDBSCAN CLUSTERS

To assess the robustness of the clustering structure identified by HDBSCAN on Level 3 (GS-DAE) embeddings, a sensitivity analysis was performed on the two main HDBSCAN parameters: *min\_cluster\_size* and *min\_samples*.

The results indicate that *min\_samples* has a negligible impact on the clustering outcome for this dataset. In contrast, *min\_cluster\_size* strongly influences both the number of detected clusters and the proportion of points classified as noise.

For values of *min\_cluster\_size* ranging from 5 to 30, the number of clusters remains relatively stable (between 25 and 30 clusters), while the silhouette score consistently exceeds 0.80 and the noise proportion remains below 2%. This behaviour suggests that the underlying clustering structure is robust and not highly sensitive to moderate parameter variations.

A marked transition occurs for *min\_cluster\_size* = 50, where the number of detected clusters decreases to 9, the proportion of noise increases to 6.9%, and the silhouette score drops to 0.57. This indicates that forcing larger clusters degrades cluster compactness and leads to the merging of previously well-separated regions of the embedding space.

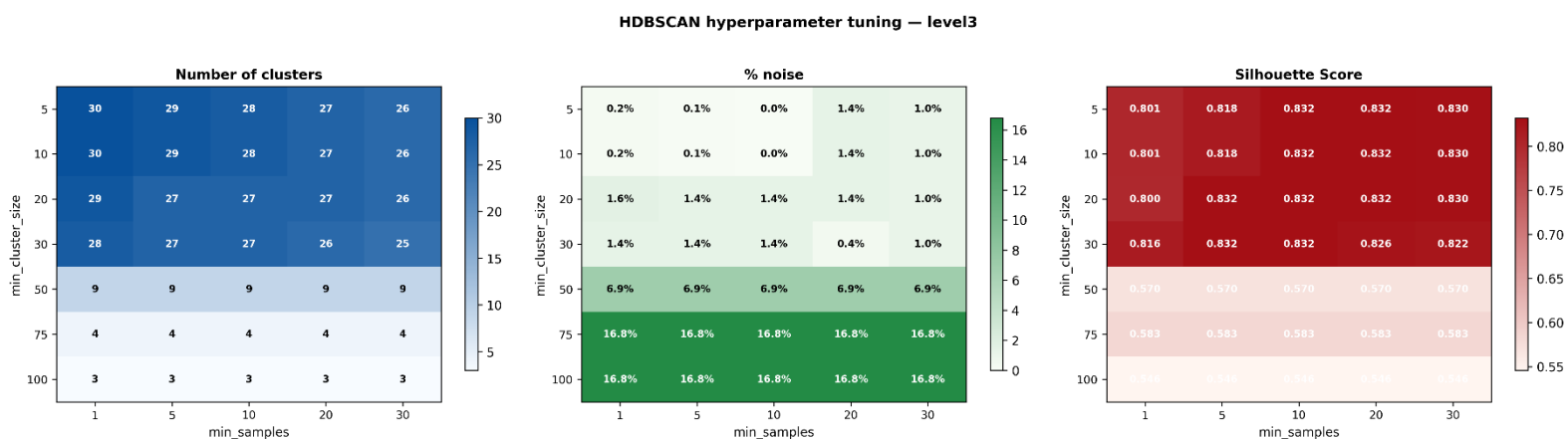


Figure 4 : The 30 clusters remain stable for *min\_cluster\_size* values between 5 and 30.

### 4.2 CLUSTER SIZE REGULARITY

An analysis of cluster sizes obtained with HDBSCAN on Level 3 (GS-DAE) reveals an unexpected regularity. Most clusters contain approximately 30 to 35 patients, resulting in a remarkably

homogeneous partition of the 953 patients into around 30 clusters, with only two patients classified as noise.

This uniform distribution of cluster sizes stands in stark contrast to the natural heterogeneity typical of biological datasets. The fact that 953 patients are perfectly divided into roughly equal-sized clusters is a strong indicator of an algorithmic bias rather than a meaningful clinical discovery. It highlights a disproportionate geometric regularisation shaping the latent space, paving the way for the oversmoothing hypothesis discussed in the following section.

### 4.3 INTERPRETATION

Level 3 (GS-DAE) embeddings are generated using a denoising autoencoder whose encoder is implemented as a Graph Neural Network operating on the protein–protein interaction network.

The denoising objective encourages the model to learn compact and robust representations by reconstructing clean inputs from corrupted observations. Simultaneously, the GNN encoder repeatedly aggregates information from neighbouring genes within the PPI graph, progressively smoothing local variations in the representation space.

The combination of these two mechanisms appears to produce highly regular latent representations characterised by numerous compact and similarly sized clusters. One possible interpretation is that graph propagation reduces local variability between patients through a phenomenon commonly referred to as *oversmoothing*, where neighbouring nodes become increasingly similar after repeated message passing operations.

Although this interpretation remains speculative, it provides a plausible explanation for both the high geometric coherence observed in Level 3 (GS-DAE) and the reduced ability of these embeddings to separate PAM50 molecular subtypes, as previously observed in Task 1.

### 4.4 VISUALISATION OF HDBSCAN CLUSTERS

To further investigate the geometry of the Level 3 (GS-DAE) embedding space, the HDBSCAN clustering solution was visualised using a two-dimensional t-SNE projection.

The projection reveals numerous compact and well-separated regions, consistent with the very high silhouette scores obtained for HDBSCAN on Level 3 (GS-DAE). The clusters appear relatively homogeneous in size and are distributed across the latent space as isolated islands rather than as a few dominant groups.

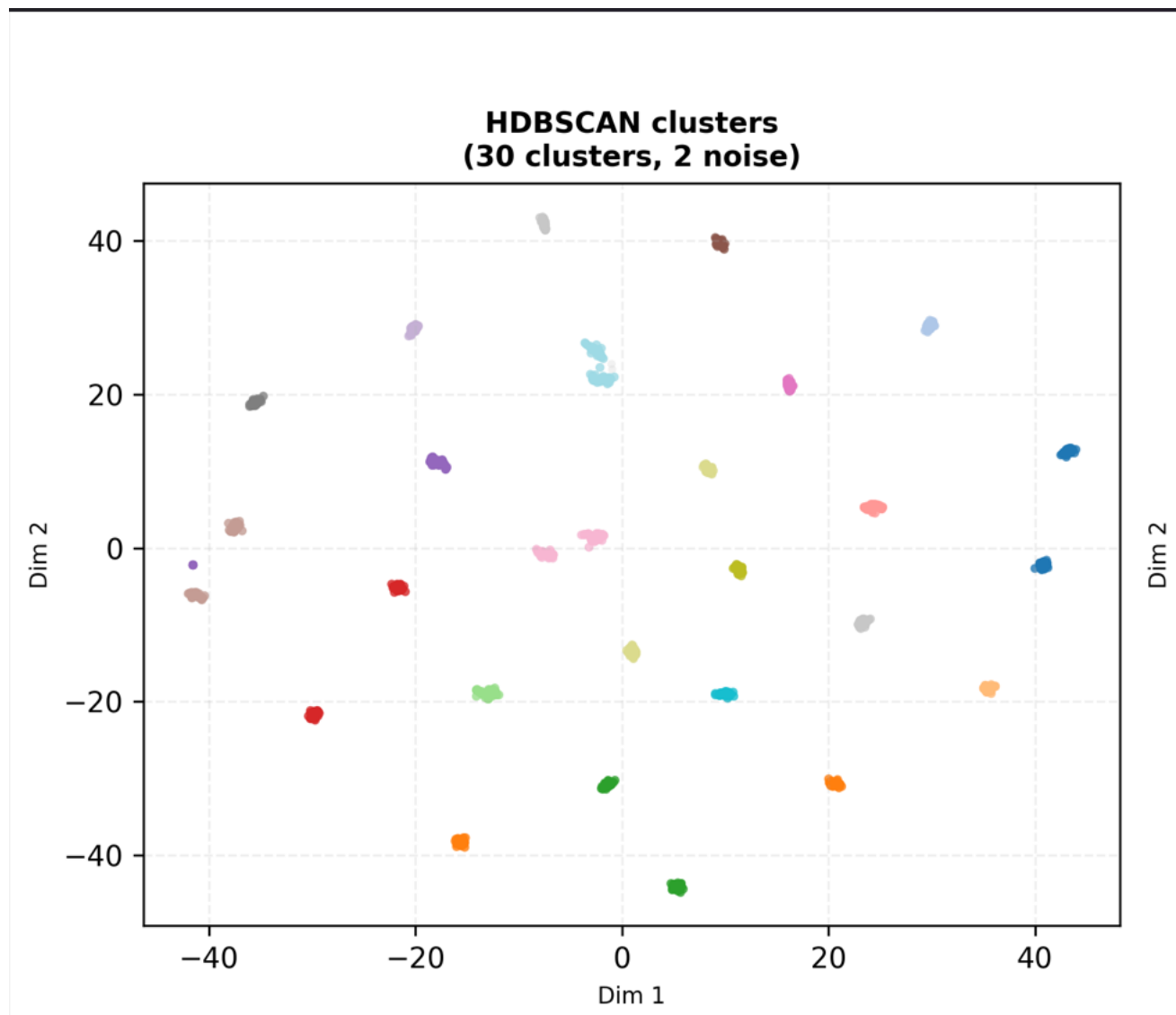


Figure 5 : t-SNE projection of HDBSCAN clusters on Level 3 embedding.

## 4.5 OVERLAYS OF CLINICAL VARIABLES

To assess whether these geometrically coherent clusters correspond to known clinical categories, PAM50 subtypes, survival groups, and pathological stages were overlaid onto the t-SNE projection.

No obvious visual separation was observed for any of these variables. Clinical labels appear distributed across multiple clusters, and no cluster is exclusively associated with a particular subtype or clinical category.

This observation is consistent with the low ARI and NMI values reported in the following section, indicating that the clustering structure identified by HDBSCAN does not directly correspond to PAM50 molecular subtypes.

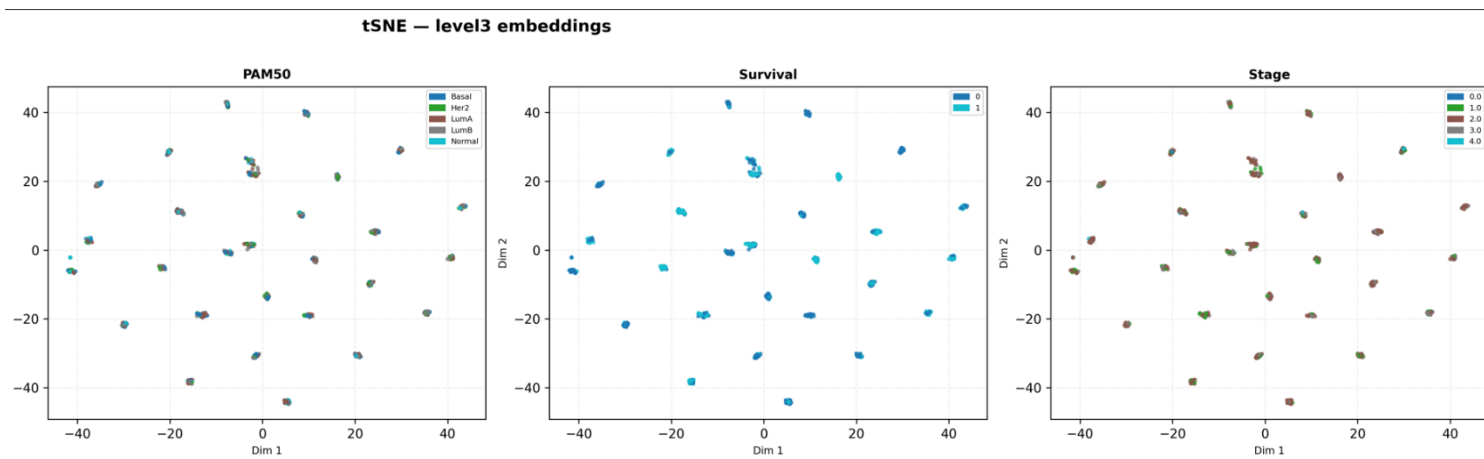


Figure 6 : The overlays indicate that clinics are uniformly distributed.

## 4.6 RELATIONSHIP WITH CLINICAL ASSOCIATIONS

The absence of visible clinical separation in the two-dimensional projection does not necessarily contradict the existence of statistically significant clinical associations.

While visual inspection primarily detects large-scale separations, chi-square tests are sensitive to more subtle enrichments of specific clinical variables within individual clusters. Consequently, weak but statistically significant associations may exist even when they are not apparent in low-dimensional visualisations.

This aspect is investigated in detail in Section 6.

## 4.7 SUMMARY OF OBSERVATIONS

Taken together, the HDBSCAN sensitivity analysis, the cluster size distribution, and the t-SNE visualisations suggest that Level 3 (GS-DAE) embeddings exhibit a highly regular and geometrically coherent structure.

This organisation is likely influenced by the combined effects of denoising and graph-based information propagation within the GNN encoder. Although these clusters do not directly correspond to established clinical categories such as PAM50, they nevertheless capture latent structures that may reflect more subtle biological patterns.

These findings suggest that Level 3 (GS-DAE) embeddings encode clinically relevant information in a distributed manner that is not directly recoverable through unsupervised clustering alone but can be effectively exploited when combined with supervised learning, as demonstrated in Task 1.

## 5 EXTERNAL METRICS

External validation metrics were used to assess whether the clusters identified in the embedding space correspond to the known PAM50 molecular subtypes of breast cancer.

For clustering algorithms requiring a predefined number of clusters (K-Means, Agglomerative Clustering, Spectral Clustering, and Gaussian Mixture Models), the common value ( $k=4$ ) selected in Section 2 was used. For HDBSCAN, the automatically determined number of clusters was retained.

For stochastic algorithms, experiments were repeated 10 times with different random seeds and results are reported as mean  $\pm$  standard deviation.

### 5.1 RESULTS

The table below summarises the Adjusted Rand Index (ARI) and Normalised Mutual Information (NMI) obtained for all algorithms and representation levels.

| Algorithm | Level  | k  | ARI_mean | ARI_std | NMI_mean | NMI_std | n_runs |
|-----------|--------|----|----------|---------|----------|---------|--------|
| KMEANS    | level0 | 4  | -0.001   | 0.0001  | 0.005    | 0.0     | 10     |
| KMEANS    | level1 | 4  | -0.0003  | 0.0007  | 0.0041   | 0.0006  | 10     |
| KMEANS    | level2 | 4  | 0.025    | 0.0001  | 0.031    | 0.0003  | 10     |
| KMEANS    | level3 | 4  | 0.003    | 0.0002  | 0.013    | 0.0003  | 10     |
| AGG       | level0 | 4  | -0.003   | 0.0     | 0.004    | 0.0     | 1      |
| AGG       | level1 | 4  | -0.001   | 0.0     | 0.004    | 0.0     | 1      |
| AGG       | level2 | 4  | 0.030    | 0.0     | 0.038    | 0.0     | 1      |
| AGG       | level3 | 4  | 0.004    | 0.0     | 0.015    | 0.0     | 1      |
| HDBSCAN   | level0 | 4  | -0.003   | 0.0     | 0.0076   | 0.0     | 1      |
| HDBSCAN   | level1 | 3  | 0.003    | 0.0     | 0.0046   | 0.0     | 1      |
| HDBSCAN   | level2 | 27 | 0.039    | 0.0     | 0.0935   | 0.0     | 1      |
| HDBSCAN   | level3 | 30 | 0.003    | 0.0     | 0.0417   | 0.0     | 1      |
| SPECTRAL  | level0 | 4  | -0.0002  | 0.003   | 0.004    | 0.0018  | 10     |
| SPECTRAL  | level1 | 4  | 0.001    | 0.0038  | 0.006    | 0.002   | 10     |
| SPECTRAL  | level2 | 4  | 0.008    | 0.007   | 0.025    | 0.0085  | 10     |
| SPECTRAL  | level3 | 4  | 0.003    | 0.0038  | 0.010    | 0.0051  | 10     |
| GMM       | level0 | 4  | -0.000   | 0.0006  | 0.003    | 0.0007  | 10     |
| GMM       | level1 | 4  | 0.0024   | 0.0012  | 0.005    | 0.0008  | 10     |
| GMM       | level2 | 4  | 0.0275   | 0.0002  | 0.035    | 0.0004  | 10     |
| GMM       | level3 | 4  | 0.003    | 0.001   | 0.014    | 0.0012  | 10     |

The corresponding heatmap is shown in Figure 7.



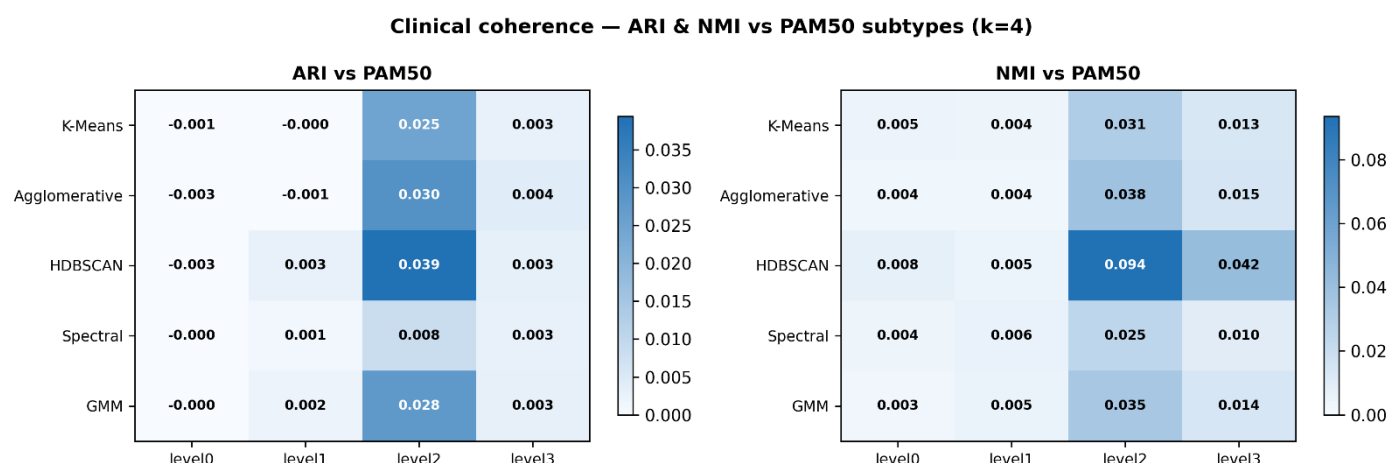


Figure 7 : The values remain close to zero across all algorithms.

Overall, ARI and NMI values remain low across all algorithms and all representation levels. No clustering method achieves a strong correspondence with the PAM50 classification.

## 5.2 INTERPRETATION

The consistently low ARI and NMI values indicate that the natural organisation of the embedding space does not directly reproduce the PAM50 taxonomy.

Several explanations may account for this observation.

First, the learned representations may capture biological information that is not reflected by PAM50 labels, such as pathway activity, mutational processes, or microenvironment-related signals.

Second, PAM50 itself represents a relatively coarse molecular categorisation and may not correspond to the dominant axes of variation present in the embedding space.

Finally, the embeddings may encode continuous biological trajectories rather than discrete patient groups, making hard clustering inherently difficult.

Interestingly, this lack of correspondence is observed even for Level 2 (GR-DAE) embeddings, which achieved the best supervised PAM50 classification performance in Task 1. This suggests that PAM50 information is present in the latent representations but is not organised as naturally separable clusters and instead requires a supervised decision boundary to be effectively recovered.

Similarly, the highly regular cluster structure identified by HDBSCAN on Level 3 (GS-DAE) does not translate into improved agreement with PAM50 labels, supporting the hypothesis that these clusters primarily reflect the geometry induced by graph propagation rather than molecular subtype organisation.

The low standard deviations observed across repeated runs further indicate that these findings are stable and reproducible across algorithms and representation levels.

### **5.3 STATISTICAL COMPARISON OF EXTERNAL METRICS**

To determine whether the relationship between cluster assignments and PAM50 labels differed significantly across representation levels, paired t-tests were performed on the ARI values obtained over the 10 repeated runs of K-Means clustering.

The comparison between Levels 2 (GR-DAE) and 3 (GS-DAE) revealed a statistically significant difference ( $p < 0.001$ ), indicating that the alignment between the latent geometry and the PAM50 taxonomy changes substantially between these two representation levels.

This result is consistent with the findings of Task 1. Level 2 (GR-DAE) embeddings, generated using graph regularisation, preserve molecular subtype information more effectively and therefore exhibit stronger agreement with PAM50 labels. In contrast, Level 3 (GS-DAE) embeddings, generated through graph propagation in the GNN encoder, appear to capture alternative biological structures that are less directly related to PAM50 classification.

Together, these results support the hypothesis that the different architectural choices emphasise distinct biological properties within the embedding space rather than converging towards a single universal representation.

## 6 CHI-SQUARE TESTS

Although the clustering solutions showed limited agreement with the PAM50 taxonomy according to ARI and NMI, they may nevertheless capture clinically relevant information that is not directly reflected by molecular subtype labels.

To investigate this possibility, chi-square tests of independence were performed between cluster assignments and the available clinical variables (PAM50, survival group, stage, T stage, N stage, and M stage).

For clustering algorithms requiring a predefined number of clusters, the main analysis was performed using the reference value ( $k=4$ ) selected in Section 2. Additional analyses using ( $k=5$ ) were conducted to assess the robustness of the observed associations with respect to the choice of ( $k$ ).

### 6.1 RESULTS

The table below summarises the statistically significant associations identified between cluster assignments and clinical variables.

| Choice of $k$ | Algorithms                              | Level            | Significant variables ( $p < 0.05$ )       |
|---------------|-----------------------------------------|------------------|--------------------------------------------|
| $K = 4$       | GMM and HDBSCAN                         | Level 3 (GS-DAE) | PAM50, Survival, T_stage, N_stage, M_stage |
| $K = 5$       | K-Means, Agglomerative, HDBSCAN and GMM | Level 3 (GS-DAE) | PAM50, Survival, T_stage, N_stage, M_stage |

Several observations emerge from these analyses.

For ( $k=4$ ), significant associations are observed almost exclusively for Level 3 (GS-DAE) embeddings. In particular, HDBSCAN and Gaussian Mixture Models exhibit statistically significant relationships ( $(p < 0.01)$ ) with PAM50, survival group, T stage, N stage, and M stage. In contrast, the pathological stage does not show significant associations with any clustering configuration.

When repeating the analysis with ( $k=5$ ), the main conclusions remain unchanged. Most clustering algorithms, with the exception of Spectral Clustering, identify significant associations for Level 3 (GS-DAE) embeddings. Importantly, the associations detected by HDBSCAN and GMM remain significant across both values of ( $k$ ), suggesting that these findings are robust and not driven by a particular choice of cluster number.

By contrast, K-Means and Agglomerative Clustering exhibit less stable results, with significant associations appearing or disappearing depending on the selected value of ( $k$ ).

While Levels 0, 1, and 2 occasionally exhibited significant associations with individual clinical variables, these associations were sparse and lacked consistency across algorithms and values of  $k$ . In contrast, Level 3 consistently exhibited multiple significant associations, particularly for PAM50, survival, and TNM variables, suggesting that clinically relevant information is more strongly structured within this representation level.

## 6.2 INTERPRETATION

These results suggest that Level 3 (GS-DAE) embeddings capture clinically meaningful information despite their weak correspondence with PAM50 molecular subtypes observed in the previous section.

The fact that HDBSCAN and GMM consistently identify significant associations with multiple clinical variables indicates that the latent structure learned by the GNN encoder reflects biologically relevant patterns that extend beyond a single clinical endpoint.

Interestingly, these associations are detected even though the corresponding t-SNE visualisations do not reveal obvious separations between clinical categories. This observation suggests that the clinical signal captured by the embeddings is subtle and distributed throughout the latent space rather than organised into clearly separated groups.

The absence of significant associations with the pathological stage deserves particular attention. Although severe class imbalance likely contributes to the difficulty of detecting stage-related patterns, the consistently poor performance observed in both the supervised analysis of Task 1 and the unsupervised analyses presented here suggests that stage information is only weakly represented within the learned embeddings. This distinction is important, as poor predictive performance may arise either from class imbalance or from a genuine lack of signal in the representation itself.

Overall, the chi-square analyses indicate that Level 3 (GS-DAE) embeddings contain clinically informative structure that is not directly recoverable through conventional clustering metrics such as ARI or NMI. Instead, the learned representations appear to encode more subtle biological and clinical relationships that become detectable through statistical association analyses and supervised learning approaches.

## 7 SPECTRAL CLUSTERING ANALYSIS

### 7.1 SPECTRAL ANALYSIS OF THE K-NN GRAPH

To better understand the geometry of the embedding spaces and the behaviour of Spectral Clustering, the normalised Laplacian of the k-nearest neighbour (k-NN) graph was analysed for each representation level.

The number of connected components was first examined. Levels 0, 1, and 2 each form a single connected component, indicating that their k-NN graphs remain globally connected. In contrast, Level 3 exhibits a highly fragmented structure composed of 28 connected components.

The eigenvalue spectrum of the Laplacian provides additional insight into the underlying organisation of the embeddings. For Levels 0, 1, and 2, visible eigengaps suggest the existence of partial spectral structure, with estimated optimal numbers of clusters around ( $k=5$ ) for Level 0, ( $k=6$ ) for Level 1, and ( $k=4$ ) for Level 2.

By contrast, the eigenvalues of Level 3 (GS-DAE) remain very close to zero (with minor negative values attributable to numerical precision errors), and no clear eigengap can be identified. This absence of spectral separation suggests that the graph structure does not provide an informative low-dimensional partition for Spectral Clustering.

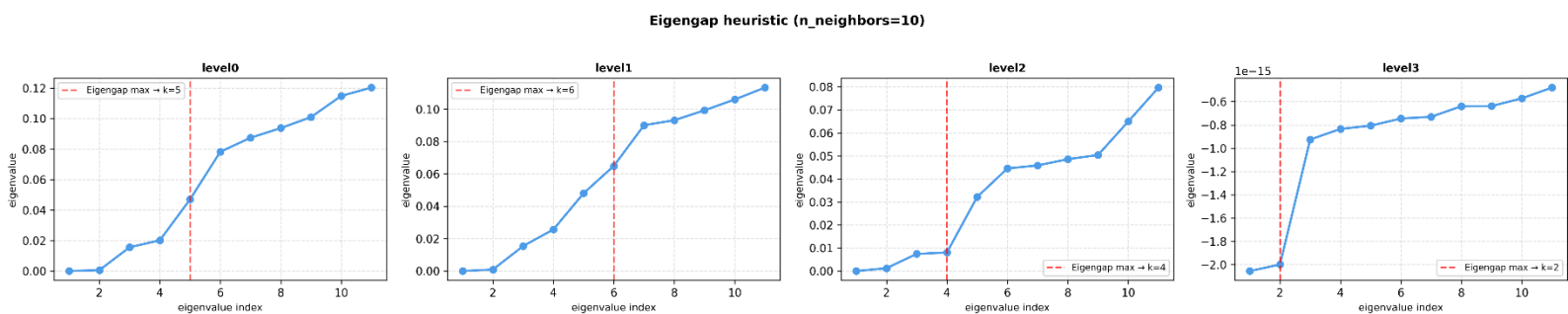


Figure 8 : Laplacian eigenvalue spectra for Levels 0 to 3.

### 7.2 CONNECTIVITY OF THE K-NN GRAPH

The neighbourhood size required to obtain a fully connected k-NN graph was also evaluated.

- Level 1 (DAE) and Level 2 (GR-DAE) require only ( $n\_neighbors=5$ );

- Level 0 (AE) requires ( $n\_neighbors=10$ );
- Level 3 (GS-DAE) requires approximately ( $n\_neighbors=100$ ), corresponding to nearly 10% of the entire cohort.

This result highlights the fragmented nature of the Level 3 (GS-DAE) embedding space. Constructing a connected similarity graph requires considering a very large number of neighbours, which contradicts the local connectivity assumption underlying Spectral Clustering.

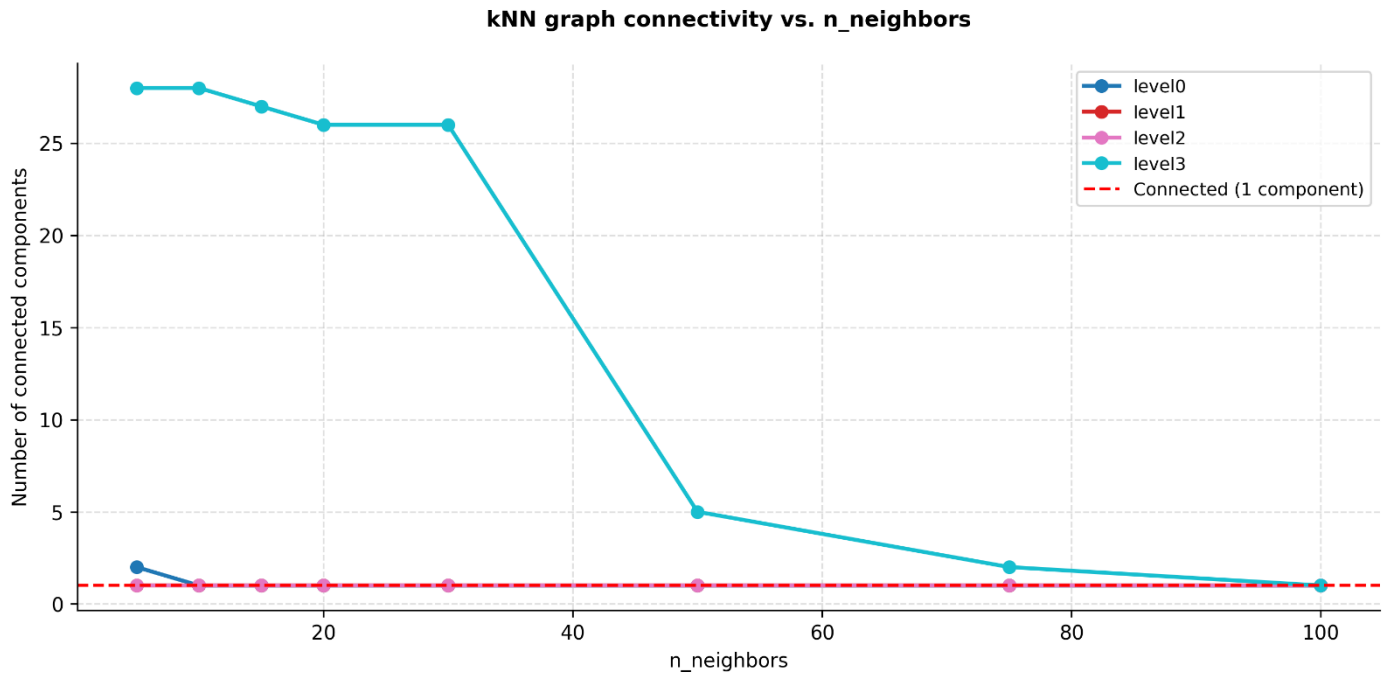


Figure 9 : Level 3 requires  $n\_neighbors = 100$  to be connected.

### 7.3 PERFORMANCE OF SPECTRAL CLUSTERING

Despite extensive tuning of the neighbourhood parameter ( $n\_neighbors$ ) ranging from 10 to 200), Spectral Clustering consistently produced poor results across all representation levels.

For Level 3 (GS-DAE) specifically:

- the similarity graph remained disconnected until approximately ( $n\_neighbors=75$ );
- silhouette scores remained negative until approximately ( $n\_neighbors=50$ ), indicating clustering solutions worse than random partitioning;
- ARI and NMI values remained close to zero for all parameter configurations.

No clear improvement trend was observed for any evaluation metric, regardless of the neighbourhood size used.

These results suggest that Spectral Clustering is poorly matched to the geometry of these embeddings, particularly for Level 3 (GS-DAE).

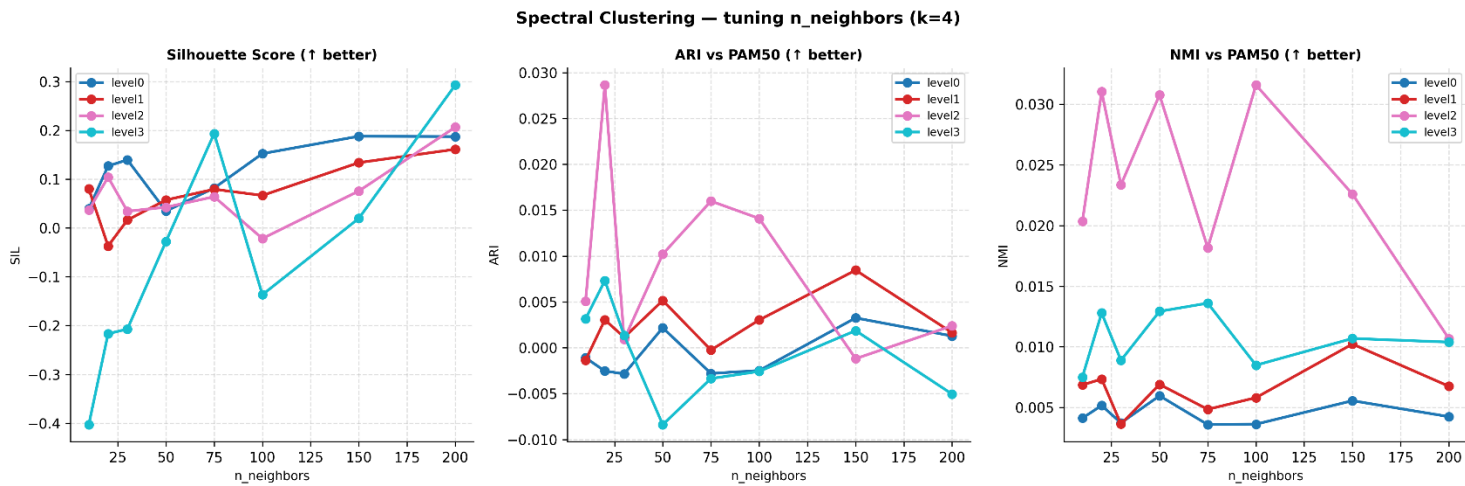


Figure 10 : The silhouette scores remain negative up to  $n\_neighbors = 50$ .

## 7.4 INTERPRETATION AND CONCLUSIONS

The previous analyses converge towards a consistent interpretation: the geometry of the learned embeddings, especially at Level 3 (GS-DAE), does not satisfy the assumptions required by Spectral Clustering.

The fragmentation of the Level 3 (GS-DAE) latent space, revealed by both the Laplacian spectrum and the connectivity analysis, prevents the construction of a meaningful local similarity graph. Consequently, the spectral embedding obtained from the graph Laplacian does not provide a useful basis for partitioning the data.

Importantly, the failure of Spectral Clustering should not be interpreted as evidence that the embeddings lack structure. On the contrary, HDBSCAN successfully identifies numerous dense and well-separated regions within the same representation space, as shown in Section 4.

This observation suggests that the geometry of Level 3 (GS-DAE) is better characterised by density-based structures than by graph-partition structures.

Similarly, the low ARI and NMI values obtained with Spectral Clustering do not imply an absence of biologically meaningful information. Rather, they indicate that the latent geometry does not align



directly with the PAM50 taxonomy and may instead capture alternative biological processes or continuous molecular gradients.

Overall, the results support three main conclusions:

- Spectral Clustering performs poorly primarily because of a mismatch between its assumptions and the geometry of the embeddings;
- the weak alignment with PAM50 reflects a geometry-versus-label mismatch rather than an absence of biological signal;
- density-based approaches such as HDBSCAN are better suited to the fragmented organisation induced by graph propagation in Level 3 (GS-DAE) embeddings.

These findings are consistent with the chi-square analyses presented in Section 6, which demonstrate that clinically relevant information remains present within the latent space despite the absence of direct correspondence with predefined clinical categories.

# CONCLUSION

## Main Findings

Task 2 investigated whether patient embeddings naturally organise into clinically meaningful groups in the absence of supervision.

The results reveal a clear dissociation between geometric coherence and alignment with established clinical taxonomies. Among the four representation levels, Level 3 (GS-DAE) embeddings consistently exhibited the strongest internal structure, particularly when analysed with HDBSCAN, indicating that the GNN-based architecture produces highly organised regions in the latent space.

However, this organisation did not translate into a direct correspondence with PAM50 molecular subtypes. External validation metrics remained low across all algorithms and levels, suggesting that the natural geometry of the embedding space reflects biological variation that differs from the supervised categories commonly used in clinical practice.

Despite this lack of alignment with PAM50, chi-square analyses revealed significant associations between Level 3 (GS-DAE) clusters and several clinically relevant variables, including survival and TNM components. These findings indicate that clinically meaningful information is present within the embeddings, although it is distributed in a subtle and non-discrete manner.

## Biological and Clinical Interpretation

The contrast between the weak PAM50 agreement and the significant clinical associations suggests that the learned representations capture broader biological processes rather than reproducing predefined molecular classifications.

From a biological perspective, graph propagation through the PPI network appears to favour the emergence of global functional patterns shared across patients. While this mechanism may attenuate fine-grained subtype-specific signatures through oversmoothing, it simultaneously enhances information related to prognosis and disease progression.

The oversmoothing effect induced by the GNN encoder appears to be the primary driver of the regular cluster structure observed in Level 3 (GS-DAE) embeddings. While it enhances global coherence and facilitates the emergence of subtle clinical associations, it also attenuates the fine-grained molecular signatures required for a direct correspondence with PAM50 subtypes.

From a clinical perspective, these findings suggest that embedding-based patient stratification may identify alternative patient groupings that are not captured by traditional classifications such as PAM50. Such representations could potentially support future applications in patient risk stratification, prognosis estimation, or treatment response prediction.

The results largely support the hypotheses formulated at the beginning of this study. Level 3 (GS-DAE) embeddings indeed exhibited the strongest internal geometric organisation without aligning with the PAM50 taxonomy, HDBSCAN proved particularly well suited to this representation level, and Spectral Clustering appeared structurally incompatible with the fragmented geometry induced by graph propagation. In contrast, the hypothesis that Level 2 (GR-DAE) would generate clinically coherent unsupervised clusters received only partial support, as improvements over Levels 0 (AE) and 1 (DAE) remained modest and did not translate into strong clinical associations.

## **Limitations and Perspectives**

Several limitations should nevertheless be acknowledged. First, the use of a fixed number of clusters for comparative purposes may obscure finer structures present in some representation levels. Second, the poor performance of Spectral Clustering highlights the importance of selecting clustering methods that are compatible with the geometry of the embedding space.

Overall, Task 2 demonstrates that incorporating biological priors through graph-based architectures substantially modifies the organisation of the latent space. While these structures do not directly reproduce existing molecular taxonomies, they encode clinically relevant information that may provide complementary insights for patient stratification.

The next stage of the project will investigate whether this information can be exploited at the individual patient level through similarity-based retrieval approaches and neighbourhood analysis.